

Logistic Regression: The Perceptron Trick

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Classification problems ([Note 3](#)); Model-based learning ([Note 6](#)); Gradient descent ([Note 57](#)); Polynomial features ([Note 61](#)).
- **Leads to:** Softmax regression ([Note 79](#)); Support vector machines ([Note 92](#)).
- **Compare with:** Support vector machines ([Note 92](#)).

1. Overview

Key point: Logistic regression separates two classes with a straight line. The perceptron trick is a simple way to find such a line: pick a random point, and if it is on the wrong side, move the line towards it.

Logistic regression is one of the most widely used classification algorithms. It is also the basic building block of neural networks: a single neuron (the **perceptron**) is very close to it. Understanding it well is a good foundation for deep learning.

Logistic regression can be explained in two ways:

- **Geometric:** find a line that separates the classes.
- **Probabilistic:** predict the probability that a point belongs to a class.

These Notes build it step by step. This Note starts with the **perceptron trick**, a very simple geometric method. It does not give the best possible line, but it explains the idea that the later Notes improve on: the sigmoid function, the loss function and gradient descent.

2. When logistic regression works

Key point: The two classes must be (almost) linearly separable: a straight line, plane or hyperplane can split them.

Take a dataset of students with two inputs, CGPA and IQ, and one output: placed or not placed. We want a model that takes a new student's CGPA and IQ and predicts whether they will be placed.

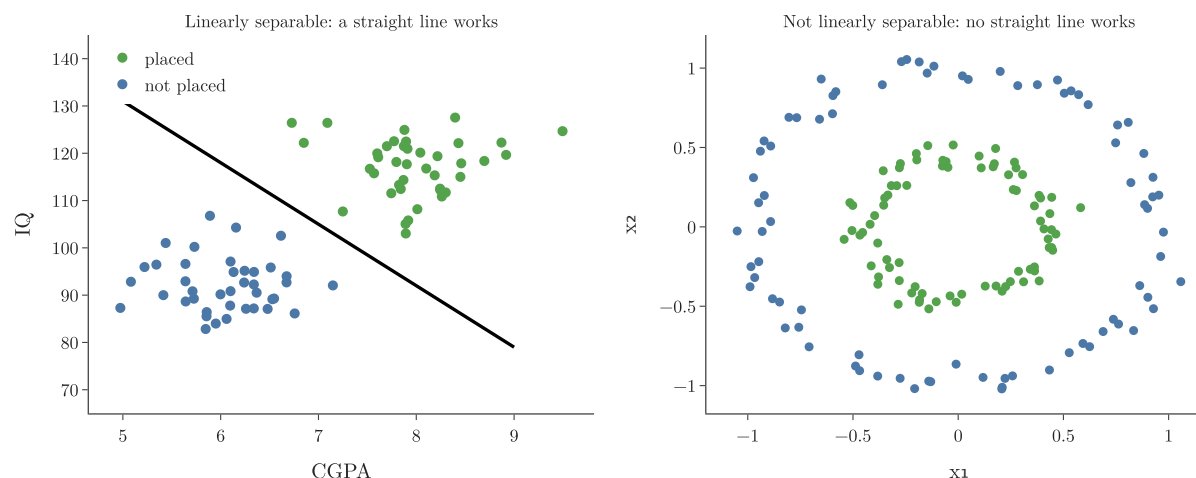


Figure 1: Linearly separable data and data that no straight line can split

Data is **linearly separable** when a straight line can split the two classes (Figure 1, left). With 3 inputs the divider is a plane, and with more inputs a **hyperplane**. A few points on the wrong side are fine: the data only needs to be almost separable.

Logistic regression draws a straight divider, just as linear regression fits a straight line. So on data like Figure 1 (right), where one class surrounds the other, it cannot do well.

3. The line and its two sides

Key point: In classification, the line is written $Ax + By + C = 0$. Putting a point into $Ax + By + C$ tells which side it is on.

3.1 The general form of a line

Key point: $Ax + By + C = 0$, or with more inputs, $w_0 + w_1x_1 + w_2x_2 + \dots = 0$.

Linear regression wrote a line as $y = mx + b$. In classification, both axes are inputs, so the line is written in its general form:

$$Ax + By + C = 0$$

With CGPA as x_1 and IQ as x_2 , this becomes $Ax_1 + Bx_2 + C = 0$. A third input adds one more term: $Ax_1 + Bx_2 + Cx_3 + D = 0$, a plane. Finding the separating line means finding good values of A , B and C .

3.2 Positive and negative sides

Key point: $Ax + By + C > 0$ on one side, < 0 on the other, $= 0$ on the line.

Every line splits the plane into a **positive side** and a **negative side**. To find which side a point (x_1, y_1) is on, put it into the left-hand side:

- $Ax_1 + By_1 + C > 0$: positive side.
- $Ax_1 + By_1 + C < 0$: negative side.
- $Ax_1 + By_1 + C = 0$: on the line.

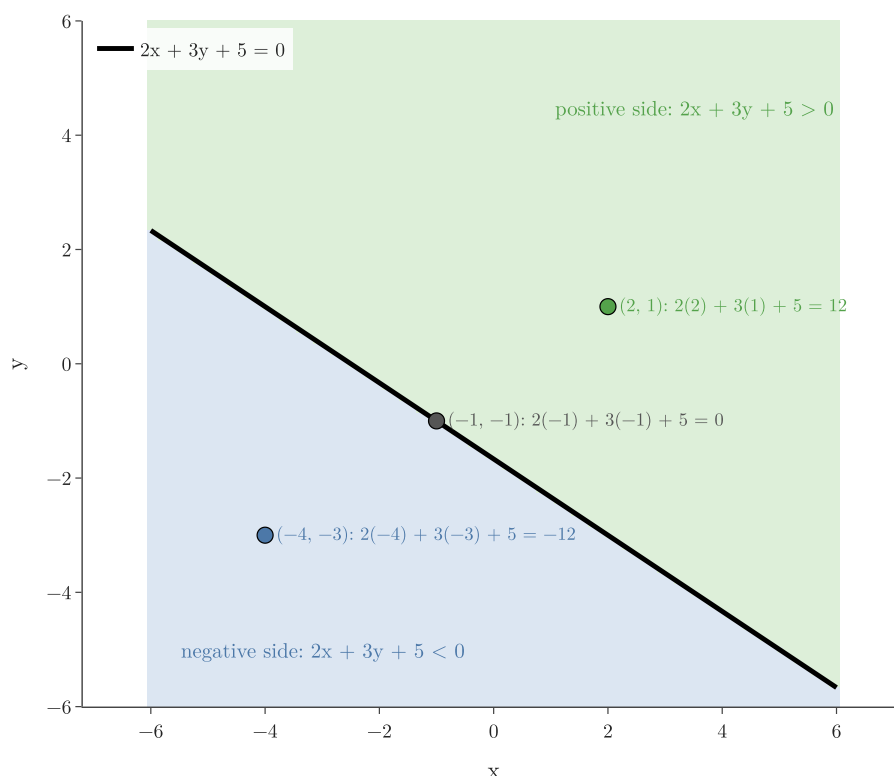


Figure 2: The line $2x + 3y + 5 = 0$ and its two sides

With numbers, for the line $2x + 3y + 5 = 0$ (Figure 2): the point $(2, 1)$ gives $4 + 3 + 5 = 12$, so it is on the positive side. The point $(-4, -3)$ gives $-8 - 9 + 5 = -12$: negative side.

4. How A, B and C move the line

Key point: Changing C shifts the line without turning it. Changing A or B turns it.

The perceptron trick moves the line by changing A , B and C . Each one has its own effect (Figure 3):

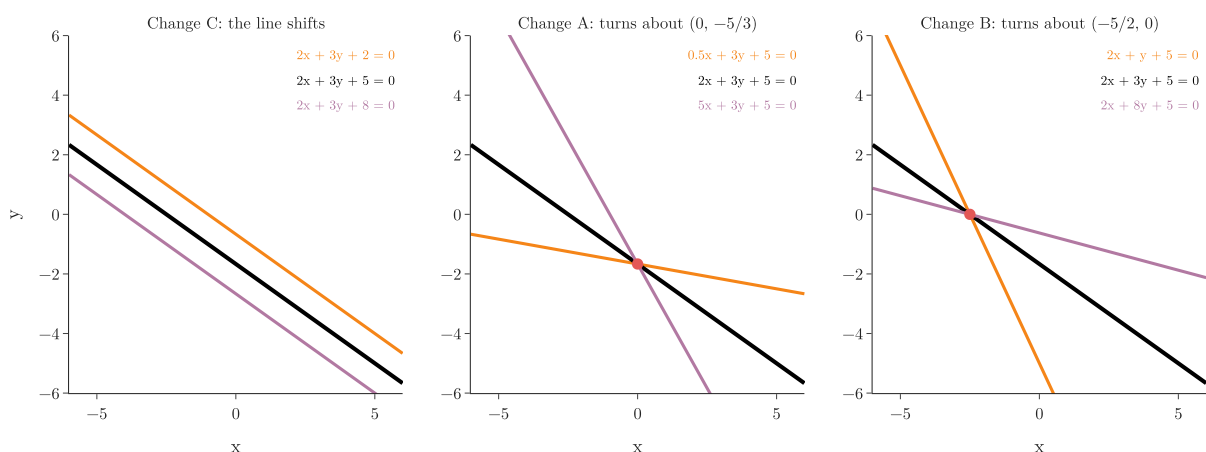


Figure 3: Changing C, A or B, one at a time

- **C:** the line moves parallel to itself. Increasing C moves $2x + 3y + C = 0$ down, decreasing it moves it up.
- **A:** the line turns about the point where it crosses the y-axis, here $(0, -5/3)$, because there $x = 0$ and A has no effect.
- **B:** the line turns about the point where it crosses the x-axis, here $(-5/2, 0)$.

Changing all three at once combines these moves, so the line can reach any position.

5. The perceptron trick

Key point: Start with any line. Repeatedly pick a random point. If it is on the correct side, do nothing; if not, move the line towards it.

The algorithm:

1. Start with random values of A , B and C : any line.
2. Repeat many times (for example 1,000):
 - pick one training point at random;
 - if it is on the correct side of the line, do nothing;
 - if it is on the wrong side, change A , B and C so that the line moves towards it.
3. Stop after the chosen number of loops, or as soon as no point is misclassified (**convergence**).

Each misclassified point “pulls” the line towards itself until it is on the correct side. After enough pulls, the line settles between the classes.

6. Moving the line towards a point

Key point: Add a 1 to the point's coordinates. For a negative point on the positive side, subtract them from (A, B, C) . For a positive point on the negative side, add them.

6.1 A negative point on the positive side

Key point: Subtract $(x, y, 1)$ from (A, B, C) .

Take the line $2x + 3y + 5 = 0$ and the point $(5, 2)$, which belongs to the negative class. It gives $10 + 6 + 5 = 21 > 0$, so it is on the positive side: misclassified.

Append a 1 to the point, $(5, 2, 1)$, and subtract it from the coefficients $(2, 3, 5)$:

$$(2 - 5, 3 - 2, 5 - 1) = (-3, 1, 4) \Rightarrow -3x + y + 4 = 0$$

Now the point gives $-15 + 2 + 4 = -9 < 0$: it is on the negative side, as it should be (Figure 4, left).

6.2 A positive point on the negative side

Key point: Add $(x, y, 1)$ to (A, B, C) .

The point $(-3, -2)$ belongs to the positive class, but gives $-6 - 6 + 5 = -7 < 0$. Add $(-3, -2, 1)$ instead:

$$(2 - 3, 3 - 2, 5 + 1) = (-1, 1, 6) \Rightarrow -x + y + 6 = 0$$

Now it gives $3 - 2 + 6 = 7 > 0$: positive side (Figure 4, right).

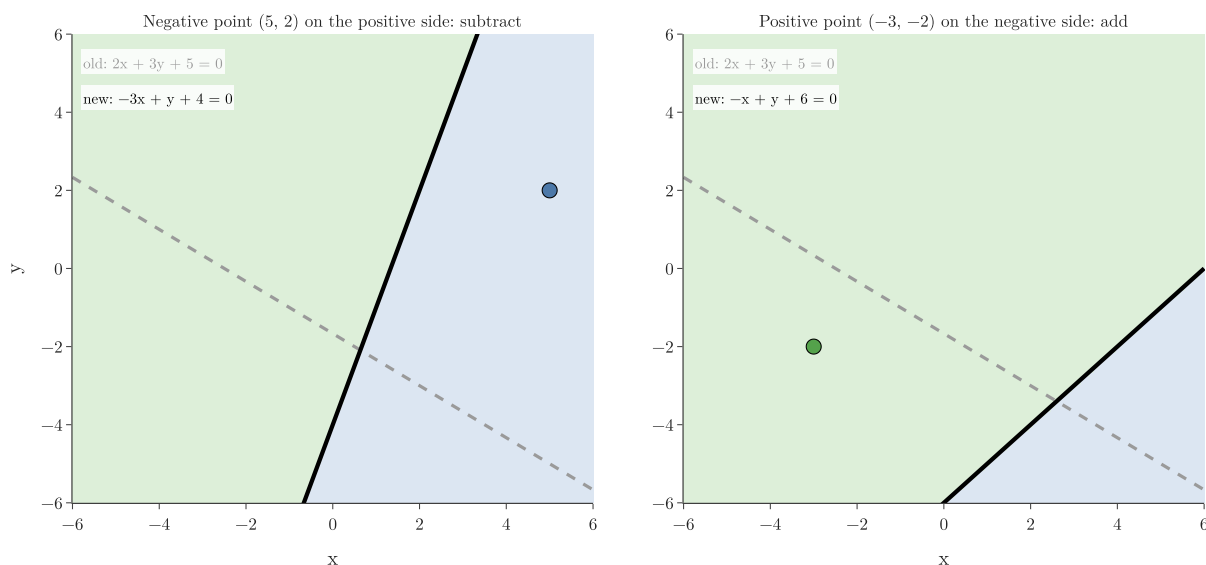


Figure 4: One update for each kind of mistake

6.3 Small steps: the learning rate

Key point: Multiply the point by a learning rate such as 0.1 before adding or subtracting, so the line moves a little at a time.

The full update above jumps the line a long way, which can undo what other points taught it. As in gradient descent, the step is scaled by a small **learning rate** η , usually around 0.1 or 0.01:

$$\text{new coefficients} = \text{old coefficients} - \eta \times (x, y, 1)$$

With $\eta = 0.1$, the first example gives $(2 - 0.5, 3 - 0.2, 5 - 0.1) = (1.5, 2.8, 4.9)$. The point's value drops from 21 to 18: the line has moved a little towards it. Repeated small steps get it to the correct side.

7. Writing the algorithm compactly

Key point: With a column of 1s and weights w , the line is $\sum w_i x_i = 0$, and all four cases fit in one update rule.

7.1 Notation

Key point: Rename C, A, B as w_0, w_1, w_2 and add a column $x_0 = 1$.

Write the line as $w_0 + w_1 x_1 + w_2 x_2 = 0$, so $w_0 = C$, $w_1 = A$ and $w_2 = B$. Then add a column x_0 that is always 1, as in multiple linear regression:

$$\sum_{i=0}^2 w_i x_i = w_0 x_0 + w_1 x_1 + w_2 x_2 = 0$$

This works for any number of inputs, just with a longer sum. In vectors it is the dot product $w \cdot x$.

7.2 Predicting

Key point: Compute $w \cdot x$ for the student. If it is positive, predict 1 (placed); otherwise predict 0.

For a student with CGPA 7.5 and IQ 110, the row is $x = (1, 7.5, 110)$. The model computes $w_0 \times 1 + w_1 \times 7.5 + w_2 \times 110$. If the result is above 0, it predicts placed (1); otherwise not placed (0). This is a **step function** of $w \cdot x$.

7.3 One update rule

Key point: $w_{\text{new}} = w_{\text{old}} + \eta (y - \hat{y}) x$. The factor $(y - \hat{y})$ is 0, +1 or -1 and picks the right action automatically.

The algorithm so far needs two checks: a negative point on the positive side, and a positive point on the negative side. Both fit into a single rule, where y is the true class and $\hat{y}$ the predicted one:

$$w_{\text{new}} = w_{\text{old}} + \eta (y - \hat{y}) x$$

True y	Predicted $\hat{y}$	$y - \hat{y}$	Update
1	1	0	none: correct
0	0	0	none: correct
1	0	+1	$w + \eta x$: add (positive point on the negative side)
0	1	-1	$w - \eta x$: subtract (negative point on the positive side)

So the loop needs no if-statements: for each random point, compute $\hat{y}$ and apply the rule.

Python: The perceptron trick.

```
def step(z):
    return 1 if z > 0 else 0

def perceptron(X, y, lr=0.1, loops=1000):
```

```
X = np.insert(X, 0, 1, axis=1)      # column of 1s
w = np.ones(X.shape[1])
for _ in range(loops):
    j = np.random.randint(0, X.shape[0])
    y_hat = step(X[j] @ w)
    w = w + lr * (y[j] - y_hat) * X[j]
return w
```

The next Note runs this on data and watches the line move.

8. Summary

- Logistic regression needs (almost) linearly separable classes.
- A line $Ax + By + C = 0$ has a positive and a negative side; plug a point in to find which.
- Perceptron trick: pick random points; move the line towards each misclassified one.
- Update: $w \leftarrow w + \eta(y - \hat{y})x$, with a column of 1s in x .
- It finds a separating line, but not necessarily the best one. The next Notes fix this.

9. Key terms

Term	Meaning
Logistic regression	A classification algorithm that separates classes with a line, plane or hyperplane
Perceptron	A single artificial neuron: a weighted sum of the inputs followed by a step
Perceptron trick	Moving a line towards each misclassified point until the classes are separated
Linearly separable	Data whose classes a straight line, plane or hyperplane can split
Hyperplane	The flat divider in more than three dimensions
Positive and negative side	The two halves of the plane where $Ax + By + C$ is above or below 0
Step function	A function that outputs 1 for positive inputs and 0 otherwise
Convergence	The point where training stops changing, here when no point is misclassified